



Lakshya JEE 2.0 (2024)

Determinants

DPP-01

1. The value of the determinant $\begin{vmatrix} 1 & x & x^3 \\ 1 & y & y^3 \\ 1 & z & z^3 \end{vmatrix}$ is equal to

- (1) $(x-y)(y-z)(z-x)$
 (2) $(x-y)(y-z)(z-x)(x+y+z)$
 (3) $(x+y+z)$
 (4) $(x-y)(y-z)(z-x)(xy+yz+zx)$

2. The value of the determinant $\begin{vmatrix} 1 & \log_b a \\ \log_a b & 1 \end{vmatrix}$ is

equal to:

- (1) 1 (2) $\log_a b$
 (3) $\log_b a$ (4) 0

3. The roots of the equation $\begin{vmatrix} x-1 & 1 & 1 \\ 1 & x-1 & 1 \\ 1 & 1 & x-1 \end{vmatrix} = 0$ are

- (1) 1, 2 (2) -1, 2
 (3) 1, -2 (4) -1, -2

4. $\begin{vmatrix} x & 4 & y+z \\ y & 4 & z+x \\ z & 4 & x+y \end{vmatrix}$ is equal to

- (1) 4 (2) $x+y+z$
 (3) xyz (4) 0

5. The value of the determinant $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix}$ is

- (1) 0 (2) 1
 (3) 2 (4) 3

6. If $\begin{vmatrix} 4 & -4 & 0 \\ a & b+4 & c \\ a & b & c+4 \end{vmatrix} = 0$, then $a+b+c$ is equal to

- (1) 41 (2) 116
 (3) 628 (4) -4

7. The roots of the equation $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0$ are:

- (1) -1, -2 (2) -1, 2
 (3) 1, -2 (4) 1, 2

8. Let $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix}$,

 then $f(100)$ is equal to

- (1) 0 (2) 1
 (3) 100 (4) -100

9. $\begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (b^x + b^{-x})^2 & (b^x - b^{-x})^2 & 1 \\ (c^x + c^{-x})^2 & (c^x - c^{-x})^2 & 1 \end{vmatrix}$ is equal to:

- (1) 0 (2) $2abc$
 (3) $a^2b^2c^2$ (4) abc

10. The determinant

$$D = \begin{vmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) & \cos 2\beta \\ \sin \alpha & \cos \alpha & \sin \beta \\ -\cos \alpha & \sin \alpha & \cos \beta \end{vmatrix}$$
 is

independent of

- (1) α (2) β
 (3) α and β (4) Neither α or β

11. If a, b, c are positive and not equal then value of

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$
 may be:

- (1) 1 (2) -3
 (3) 2 (4) 4



12. The value of θ satisfying the equation

$$\begin{vmatrix} 1 + \sin^2 \theta & \cos^2 \theta & 4 \sin 4\theta \\ \sin^2 \theta & 1 + \cos^2 \theta & 4 \sin 4\theta \\ \sin^2 \theta & \cos^2 \theta & 1 + 4 \sin 4\theta \end{vmatrix} = 0$$
 are:

- (1) $\frac{4\pi}{24}, \frac{3\pi}{24}$ (2) $\frac{7\pi}{24}, \frac{5\pi}{24}$
 (3) $\frac{7\pi}{24}, \frac{11\pi}{24}$ (4) $\frac{\pi}{24}, \frac{11\pi}{24}$

13. If $A + B + C = \pi$, then value of

$$\begin{vmatrix} \sin(A+B+C) & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ \cos(A+B) & -\tan A & 0 \end{vmatrix}$$
 is

- (1) 0 (2) 1
 (3) $2 \sin B \cdot \tan A \cos C$ (4) $2 \sin A \sin B \cdot \sin C$

14. If ω is non-real complex cube root of unity, then

$$\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & \omega & 1 \end{vmatrix}$$
 is equal to

- (1) 0 (2) 1
 (3) 3 (4) 2



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (2)
2. (4)
3. (2)
4. (4)
5. (1)
6. (4)
7. (2)

8. (1)
9. (1)
10. (1)
11. (2)
12. (3)
13. (1)
14. (1)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>